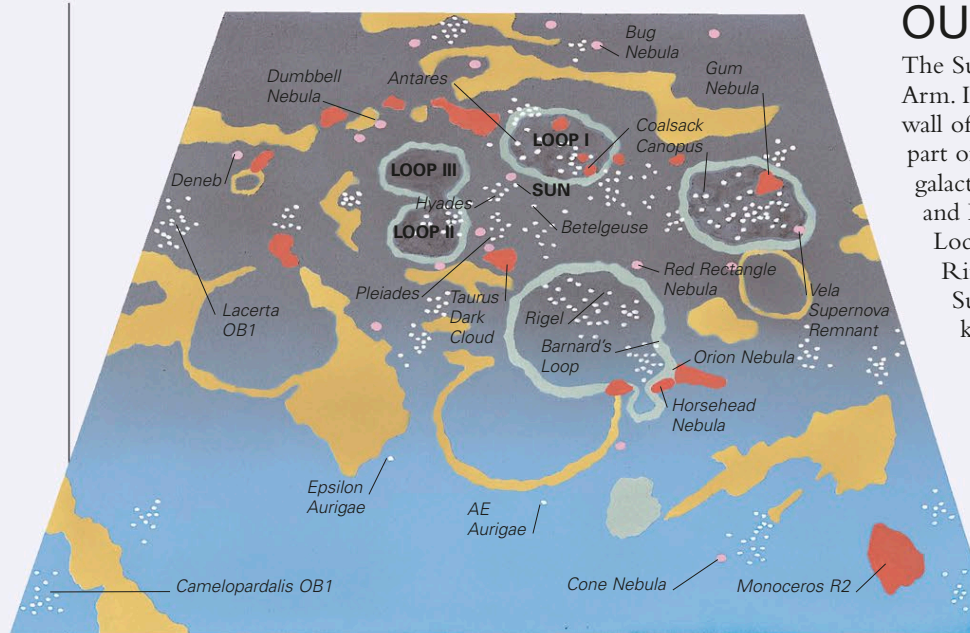


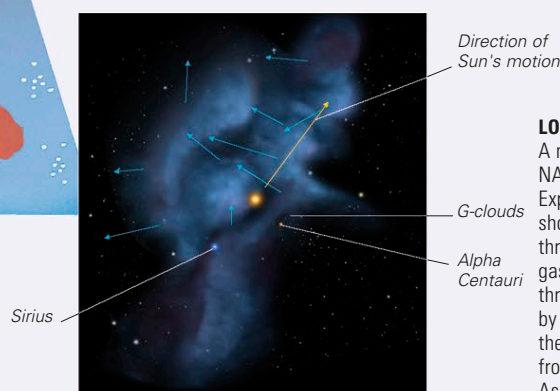


REGIONAL MAP

This schematic representation of the solar system's local neighborhood maps out a section of the Milky Way's Orion Arm about 5,000 light-years across. The Sun is located just above center. Hydrogen gas clouds are marked in brown, molecular clouds are in red, and interstellar bubbles are colored green. Nebulae are shown in pink, while star clusters and giant stars are featured in white.

OUR LOCAL NEIGHBORHOOD

The Sun lies in one of the less-dense regions of the Milky Way's Orion Arm. It sits in a "bubble" of hot, ionized hydrogen gas bounded by a wall of colder and denser neutral hydrogen gas. The Local Bubble is part of a tubelike chimney that extends through the disk into the galactic halo. The largest local coherent structure, detected by radio- and X-rays, is known as Loop I. This is believed to be part of the Local Bubble impacting into a molecular cloud known as the Aquila Rift. Two other expanding bubbles, Loops II and III, lie nearby. The Sun is traveling through material flowing out from the young stars known as the Scorpius–Centaurus Association, toward the Local Interstellar Cloud, a mass of dense interstellar gas.

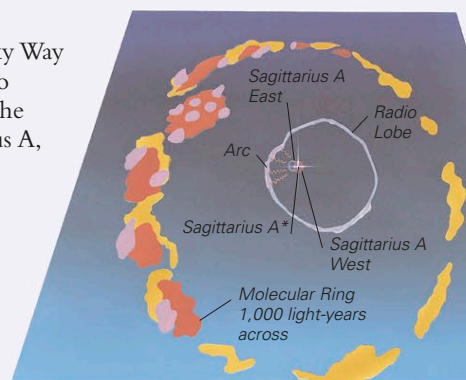


LOCAL BUBBLE

A map of local space from NASA's Interstellar Boundary Explorer (IBEX) satellite shows the Sun's motion through tenuous clouds of gas whose own motions through space are indicated by blue arrows. In general, the gas is blowing away from the Scorpius–Centaurus Association of young stars.

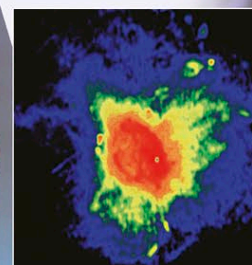
THE GALACTIC CENTER

Dense layers of dust and gas obscure the center of the Milky Way from us in optical wavelengths. However, the brightest radio source in the sky is located toward the Galactic Center in the constellation of Sagittarius. This source, known as Sagittarius A, consists of two parts. Sagittarius A East is believed to be a bubble of ionized gas, possibly a supernova remnant. Sagittarius A West is a cloud of hot gas, and embedded within it is a very strong and compact radio source, called Sagittarius A* (Sgr A*). Sgr A* appears to have no orbital motion and therefore probably lies at the very center of the Milky Way. It has a radius of less than 1.4 billion miles (2.2 billion km)—smaller than that of Saturn's orbit—and orbital motions of the gas clouds around it indicate that it surrounds a supermassive black hole of about 3 million solar masses. Centered on Sgr A* is a three-pronged mini-spiral of hot gas, about 10 light-years in diameter, and surrounding this is a disk of cooler gas and dust called the Circumnuclear Disk.



GALACTIC CENTER

Surrounding Sagittarius A, the Radio Lobe is a region of magnetized gas including an arc of twisted gas filaments. Farther out, the expanding Molecular Ring consists of a series of huge molecular clouds (red) and an association of hydrogen clouds (brown) and nebulae (pink). The two smaller gas disks around Sagittarius A cannot be seen at this scale.



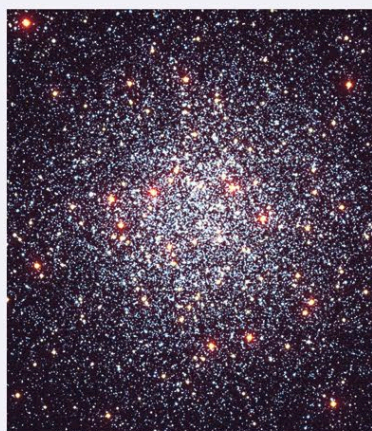
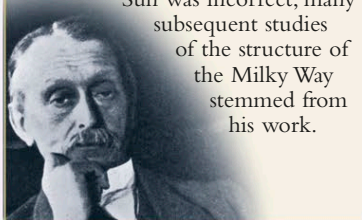
RADIO MAPS

Radio maps of Sagittarius A show a spiral pattern of hot, ionized gas that appears to be falling into the very center of the Milky Way. Situated at the middle of the maps is the point source Sagittarius A*, thought to be a supermassive black hole at the very heart of the Milky Way.

J. C. KAPTEYN

Dutch astronomer Jacobus Cornelius Kapteyn (1851–1922) was fascinated by the structure of the Milky Way. Studying at the University of Groningen, he used photography to plot star densities. He arrived at a lens-shaped galaxy with the Sun near its center.

Although his positioning of the Sun was incorrect, many subsequent studies of the structure of the Milky Way stemmed from his work.



GLOBULAR CLUSTER

Like bees around a honey pot, the stars of a globular cluster swarm in a compact sphere. Containing up to a million (mostly Population II) stars, most of these clusters are found in the Milky Way's halo.

THE EDGES OF THE MILKY WAY

Surrounding the disk and central bulge of the Milky Way is the spherical halo, stretching out to a diameter of more than 100,000 light-years. Compared to the density of the disk and the bulge, the density of the halo is very low, and it decreases as it extends away from the disk. Throughout the halo are about 200 globular clusters (see pp.288–289), spherical concentrations of older, Population II stars (see p.227). Individual Population II stars also exist in the halo. These halo stars orbit the galactic center in paths that take them far from the galactic disk, and because they do not follow the motion of the majority of the stars in the disk, their relative motion to the Sun is high. For this reason, they are sometimes called high-velocity stars. Calculations of the mass of the Milky Way suggest that 90 percent consists of mysterious dark matter (see p.27). Some of this may be composed of objects with low luminosities, such as brown dwarfs and black holes, but most is believed to be composed of exotic particles, the nature of which have yet to be discovered. The halo extends into the corona, which reaches out to encompass the Magellanic Clouds (see pp.310–311), the Milky Way's nearby neighbors in space.